PCCOE International Grand Challenge: 2026
Pune, India

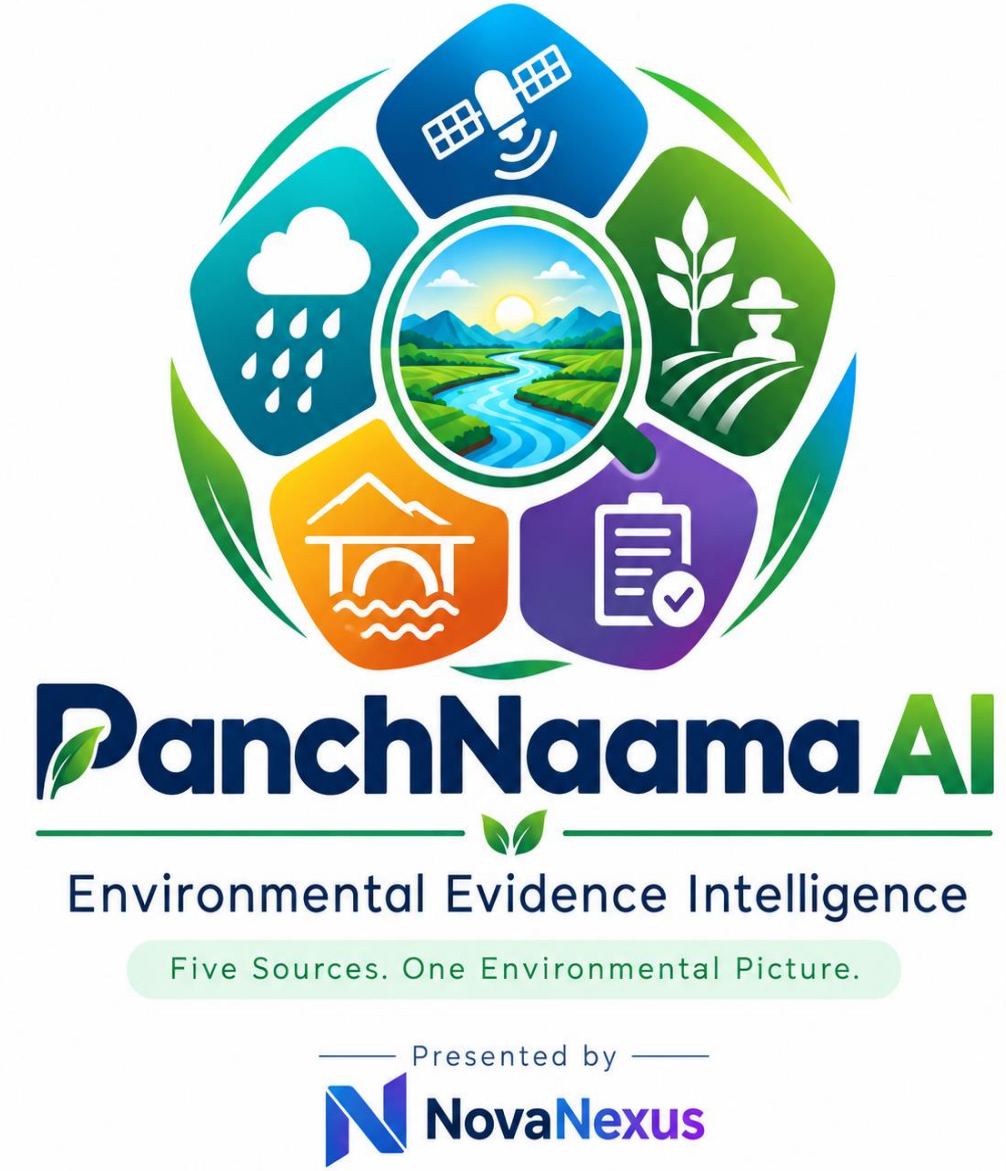
Theme: AI for Climate Change

PANCHNAAMA AI

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Outline

- Problem Statement/Idea
- Objectives
- Proposed Solution/Architecture
- Unique features/Innovation
- Expected Outcomes
- Conclusion



Problem Statement

The Problem

After a flood, damage is assessed manually, plot by plot or village by village — often weeks after the water recedes. Two similarly affected areas can end up in very different compensation categories even when the underlying damage looks the same, because there is no independent way to check assessments against real environmental evidence

Why it matters

1, Compensation is livelihood-critical for flood-hit farming families

2. Assessments can lag weeks behind the actual flood event

3. Different agencies have reported conflicting damage figures for the same flood

CURRENT GAPS IN EXISTING APPROACHES

Existing Approach	Key Limitation
Manual field survey / panchnama	Slow, inconsistent across surveyors, hard to audit at scale
Aggregate government disbursement data	Published at district level only — masks village/plot-level unfairness
Farmer grievance & appeal process	Reactive; burden of proof falls on the farmer, with no independent evidence

OUR IDEA

Reconciliation AI cross-checks flood-damage assessments against satellite, rainfall and terrain evidence — flagging where the assessment doesn't match reality, and investigating why.

Objectives

1 Detect

Flag mismatches between the official assessment and independent evidence.

2 Explain

Surface plausible causes — drainage, terrain, timing — behind each mismatch.

3 Prioritise

Rank cases by confidence so investigators know where to look first.

4 Support

Assist human review — the system never auto-decides compensation.

EXPECTED IMPACT

Efficiency	A ranked shortlist replaces blind manual review of every case
Fairness	Similar damage gets similar treatment, regardless of who assessed it or when
Transparency	Every flag ships with evidence and a confidence score, not a black-box verdict
Long-term resilience	Historical drift tracing shows where flood risk is worsening over time

WHO BENEFITS

1. Flood-hit farmers & communities
2. District disaster-management authorities
3. State revenue & agriculture departments
4. Insurance and releif agencies

ONE RULE ACROSS EVERY OBJECTIVE

The system flags and explains evidence-based mismatches — it never decides compensation.

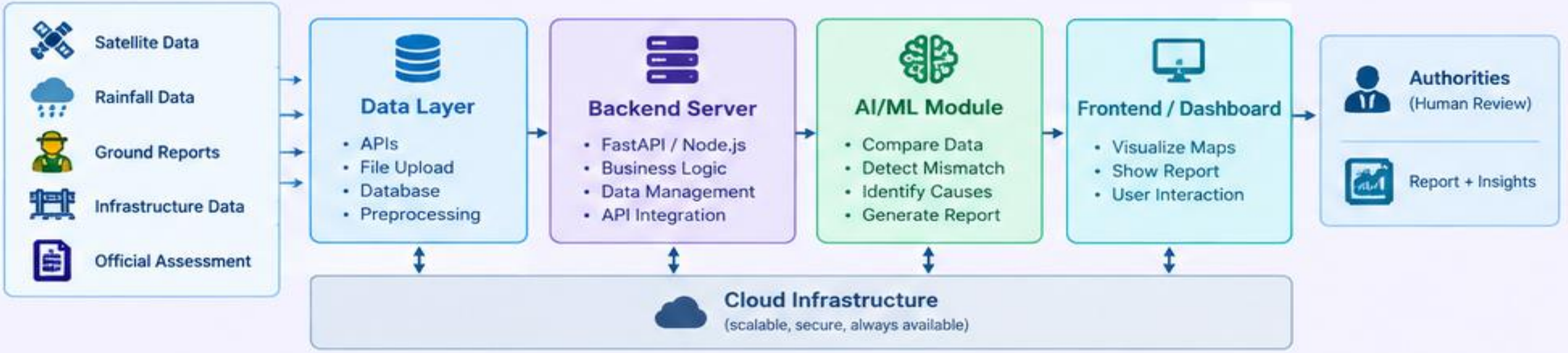
Proposed Solution & Architecture

Easy to Scale
- Works for small or large areas.
- Cloud-based system can handle more data.

Easy to Expand
- Can cover more districts over time.
- More data sources can be added later.

Practical & Affordable
- Uses available satellite and rainfall data.
- Uses open-source tools to reduce cost.

Python
OpenCV
GeoPandas
PyTorch
NumPy
Matplotlib
scikit-learn
Jupyter
Google Cloud
FastAPI
React
Leaflet
PostgreSQL
Node.js
Docker
Git



Unique features & Innovation

- What makes Reconcilation AI different from existing solutions?

Multi-Source Data
Combines satellite, rainfall, ground reports, infrastructure and official data in one system.

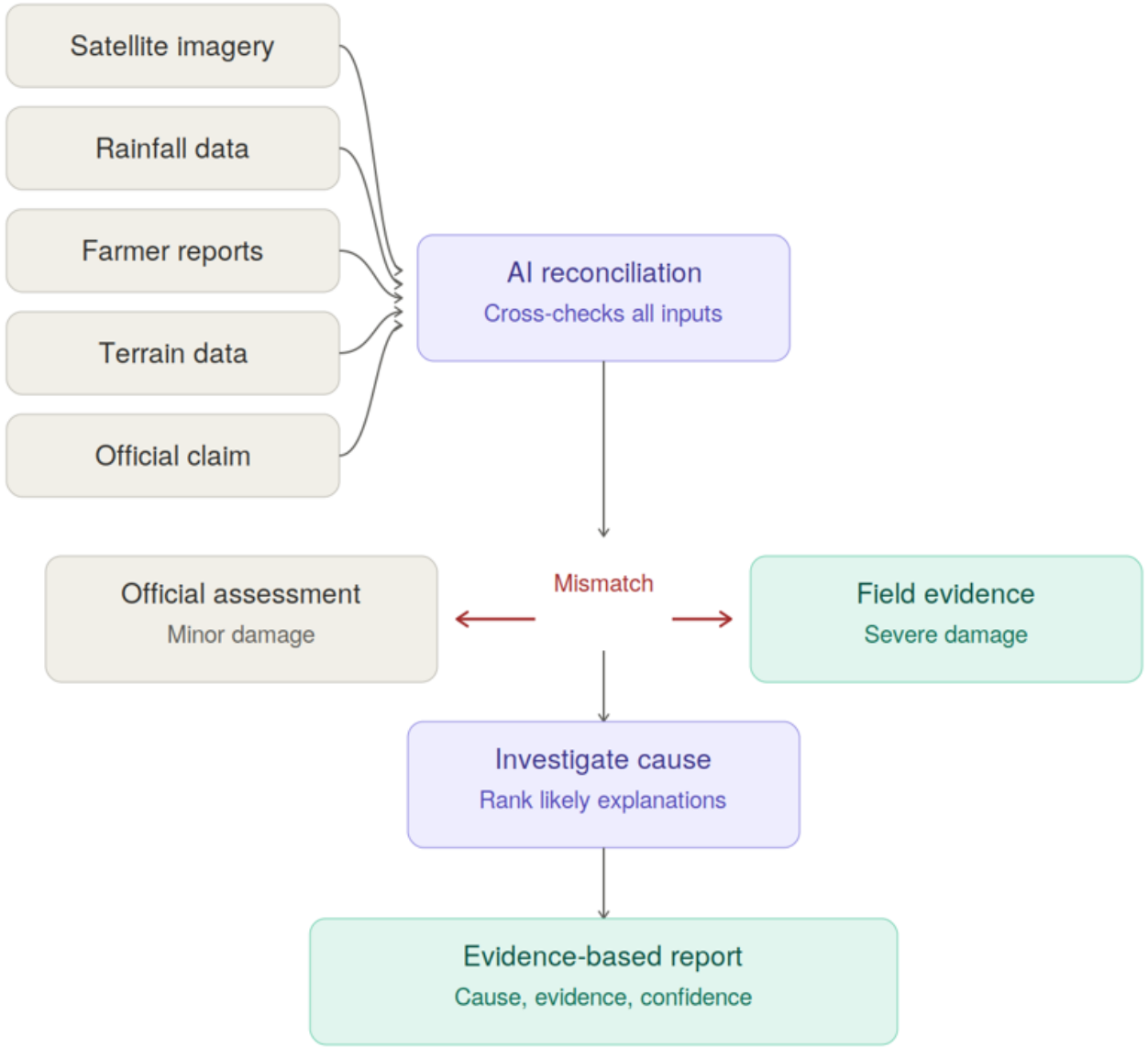
Mismatch Detection
Compares real evidence with the official damage assessment and flags possible differences.

Cause Investigation
Uses data and historical changes to find possible reasons for the mismatch, such as drainage or terrain.

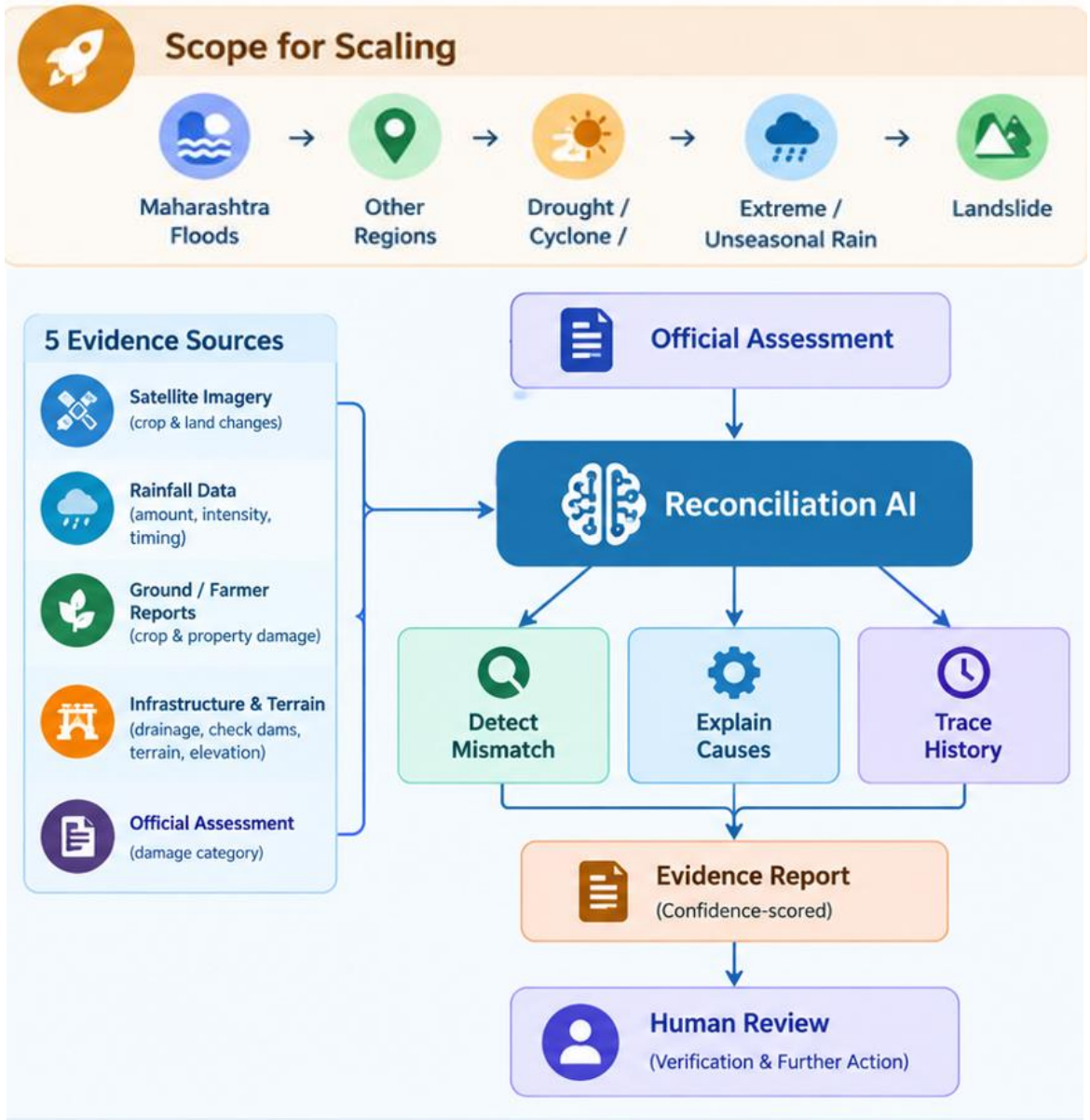
Evidence-Based Report
Generates a simple report with evidence, possible cause, timeline and confidence level for human review.

Our innovation: the mismatch detection layer

Flags gaps between what's reported and what the data shows



Expected Outcome



📦 **Deliverables:**

- **Reconciliation AI Dashboard** — map-based evidence & flagged cases
- **Evidence Comparison** — official assessment vs. 5 evidence sources
- **Reconciliation Report** — mismatch, possible causes, history & confidence
- **Round 2 Prototype** — real-data validation in a selected Maharashtra flood-affected region

⚡ **Short-Term Impact:**

- Prioritize cases requiring human verification
- Unify 5 evidence sources in one investigation workflow
- Reveal local variations in flood damage
- Support faster, evidence-based re-verification

🌱 **Long-Term Impact:**

- Build a historical, geo-tagged evidence archive
- Identify recurring flood-risk & infrastructure gaps
- Complement existing satellite-based assessment systems
- Support better disaster-management planning

Conclusion

💡 Our Idea

Reconciliation AI cross-checks official flood-damage assessments with satellite, rainfall, ground and infrastructure evidence. It detects potential mismatches, explains possible causes and traces their history — providing evidence for human verification.

ASSESS → CROSS-CHECK → RECONCILE → EXPLAIN → VERIFY

Why Should This Idea Be Selected?

🧠 **Explainable AI**
Shows what differs, when it occurred and possible causes — not just a mismatch flag.

👤 **Human-in-the-Loop**
AI supports verification final decisions remain with authorized authorities.

Multi-Source Evidence
Combines 5 complementary evidence layers instead of relying on one source.

🚀 **Scalable Solution**
Maharashtra Floods → Other Regions → Multi-Hazard Applications

We don't replace the assessment — we add an evidence layer to help verify it.